

Day 10

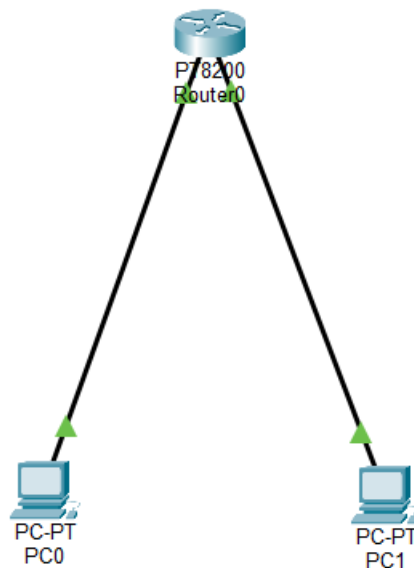
Review and Practice Practical (1 hour): Create a physical network topology in Cisco Packet Tracer. **Task:** Document the steps for configuring a small network.

1. Open Cisco Packet Tracer and create a new project file named "HomeWiFiNetwork.pkt" with three PCs and one wireless router connected appropriately.

ANS :

- **Steps**

1. Open Cisco Packet Tracer.
2. Create a new project.
3. Drag and drop one **Wireless Router** into the workspace.
4. Drag and drop **three PCs** into the workspace.
5. Connect each PC to the wireless router using Copper Straight-Through cables.
6. Save the project as **HomeWiFiNetwork.pkt**.



2. Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

ANS :

Router Configuration

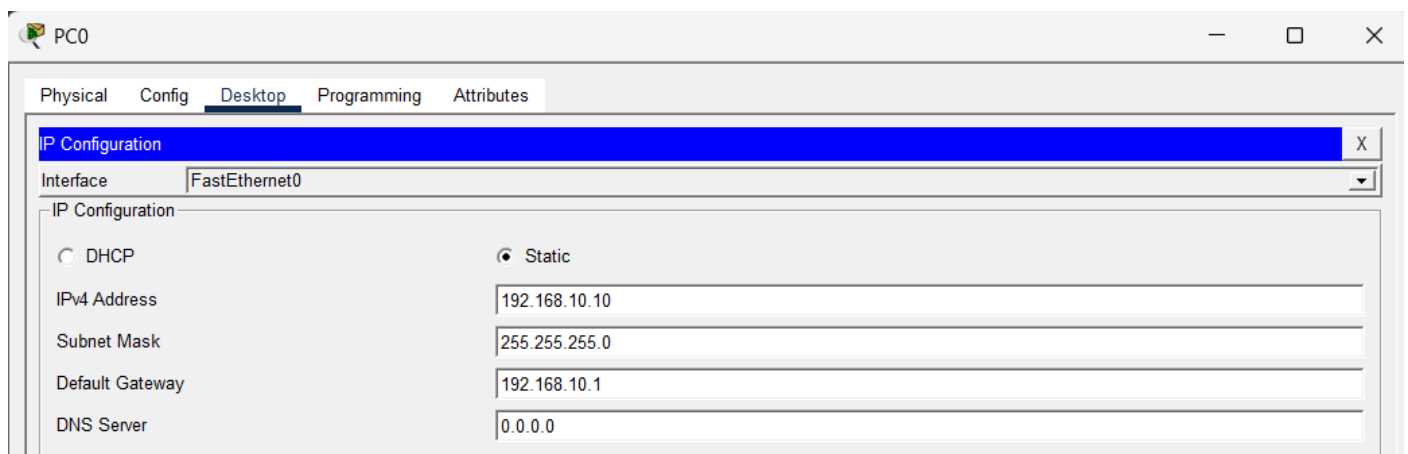
Device	IP Address
Wireless Router	192.168.10.1

PC IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.10	255.255.255.0	192.168.10.1
PC2	192.168.10.20	255.255.255.0	192.168.10.1
PC3	192.168.10.30	255.255.255.0	192.168.10.1

Steps

1. Click on PC1.
2. Go to **Desktop** → **IP Configuration**.
3. Enter:
 - IP Address: 192.168.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
4. Repeat for PC2 and PC3 using their assigned IP addresses.



3. Test connectivity by using the Packet Tracer ping tool to verify that all PCs can communicate with each other and the router.

ANS :

Commands Used

From PC1

ping 192.168.10.20

ping 192.168.10.30

ping 192.168.10.1

From PC2

ping 192.168.10.10

ping 192.168.10.30

ping 192.168.10.1

From PC3

ping 192.168.10.10

ping 192.168.10.20

ping 192.168.10.1

Sample Result

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Reply from 192.168.10.30: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Conclusion

All PCs successfully communicated with each other and with the router, confirming proper network configuration.

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

ANS :

Step 1: Create the Network

- Open Cisco Packet Tracer.
- Add one Wireless Router.
- Add three PCs.
- Connect all PCs to the router.

Step 2: Configure Router

- Open the router configuration.
- Assign IP address 192.168.10.1.

Step 3: Configure PC1

- Desktop → IP Configuration.
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 4: Configure PC2

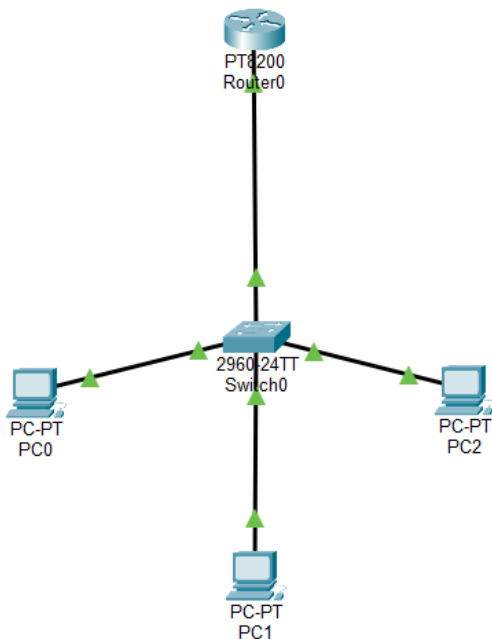
- Desktop → IP Configuration.
- IP Address: 192.168.10.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 5: Configure PC3

- Desktop → IP Configuration.
- IP Address: 192.168.10.30
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 6: Verify Connectivity

- Open Command Prompt on each PC.
- Use the ping command to test communication.
- Confirm successful replies from all devices.



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.30
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0